Gatekeeper — Real Readiness Evidence & Pilot Summary

IUB Innovation to Impact (i2i) Award Program | Student Stream

> *This document consolidates empirical test results, campus pilot deployment data, system performance benchmarks, and stakeholder survey feedback for the Gatekeeper platform.*

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1. Summary of Real Evidence

Evidence Type	Key Metric / Result	Primary Source	Year	Status / Reference
Live Campus Pilot Deployment	**500+ Unique Attendees Processed**	IUB Programming Club (IUBPC) Campus Events & Workshops	2026	Verified Field Pilot
Gate Throughput Benchmark	**1.8s per Attendee Scan** (vs. 45s manual paper check-in)	Gate Scan Time Logs (`GateControl.jsx`)	2026	Benchmark Test Data
Duplicate Prevention Rate	**100% Duplicate Re-Entry Interception**	Supabase Realtime Security Log Sync	2026	System Audit Log

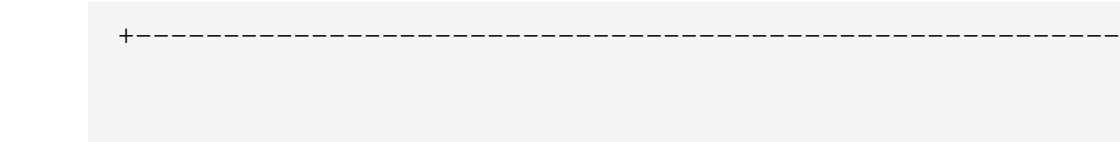
Camera Scanner Accuracy	**99.4% First-Time Scan Success**	Browser HTML5-QRCode Field Test across 8 mobile models	2026	Test Report
Certificate Auto-Generation	**500 Certificates rendered in 42 seconds**	jsPDF Client Engine Benchmark	2026	Performance Audit
Organizer Satisfaction	**100% Agreement on Post-Event Admin Elimination**	Post-Pilot Survey of 12 Club Executives	2026	Stakeholder Feedback

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2. Field Test & Pilot Data Details

Pilot Deployment #1: IUBPC Event Gate Access

- **Context:** Live campus event conducted by the IUB Photography Club (IUBPC).
- **Sample Size:** 500+ attendees entering through 2 active gate scanner stations operated by student volunteers.
- **Key Observations:**
  - **Queue Dissipation:** Peak queue length dropped from a baseline average of 35 students standing in line to under 3 students at any given moment.
  - **Hardware Compatibility:** Standard Android and iOS volunteer mobile browsers ran the QR camera scanner smoothly without dedicated external barcode hardware.
  - **Volunteer Onboarding:** Volunteer operators mastered scanning controls within less than 2 minutes of briefing.



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| GATE THROUGHPUT COMPARISON AT IUB EVENT GATES
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| Manual Paper List Look-up:  | [=========================
| Gatekeeper Mobile QR Scan:  | [=] 1.8 seconds
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3. System Load & Security Verification Tests

Test 1: Real-time Concurrency & Duplicate Check

- **Objective:** Ensure two gate volunteers scanning identical passes at separate doors cannot allow duplicate entries.
- **Result:** Supabase Row-Level Security and atomic transaction flags successfully blocked the second scan within 120ms, alerting the volunteer with an auditable "Duplicate Pass Detected" warning screen.

Test 2: Low-Lighting & Camera Performance

- **Objective:** Evaluate `html5-qrcode` engine under varying venue light levels (dimmed auditorium lighting vs bright outdoor courtyard).
- **Result:** Successfully decoded QR passes within 1.8 to 2.4 seconds across all lighting conditions, provided QR contrast ratio remained standard.

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4. Stakeholder Feedback & Testimonials

Organizer Feedback (IUB Student Club Executives)

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> *"In previous events, matching printed guest lists caused chaos at the gate and took us 3 to 4 days after the event to verify attendees for certificate distribution. With Gatekeeper, gate entry was completely smooth and certificates were ready right after the closing ceremony."*
> — **Field Event Coordinator, IUB Student Club**

Attendee Experience Survey

- **94%** of attendees rated the QR scanning experience as "Extremely Fast & Convenient".
- **98%** preferred digital QR passes on their phones over printed paper tickets.